LESSON PLAN · SCRATCH JUNIOR (K–2) INTERMEDIATE: STORY MAKERS

Lesson 10: Two-Bug Detective

Module 4 | 30 minutes | Debugging step by step | KG–Grade 2 · ages 5–7

Objective: Find and fix more than one bug in a script by testing after each fix.

Success criteria

- I can find the first bug.
- I test after each fix.
- I keep going until it works.

Key vocabulary

- **detective**: Someone who finds clues.
- **test**: Run it to check.
- **persist**: Keep trying.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (30 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Spot the difference (5 min): two pictures with two differences.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Squash two bugs: learners follow the steps.
5 min	**Perform**	Lost duck, Grumpy goodbye, Keep going
3 min	**Probe**	Busy bee
4 min	**Wrap-up**	Exit ticket: What do you do after fixing a bug?

Unplugged warm-up

Spot the difference (5 min): two pictures with two differences.

Teaching the idea

Systematic debugging with multiple faults: locate the first divergence, fix, re-test, repeat. Builds persistence.

Analogy: A detective solving a case one clue at a time.

Live demo: Narrate fixing two bugs, testing after each.

Questions to ask

Ask	Listen for
What went wrong first?	Find it.
Did that fix everything?	Test again.
How do you feel when you fix a bug?	Proud!

Differentiation

Support
Tell the child how many bugs there are.

Stretch
Make a two-bug puzzle for a partner.

Common misconceptions

Misconception	How to fix it
One fix fixes everything	There can be more bugs.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

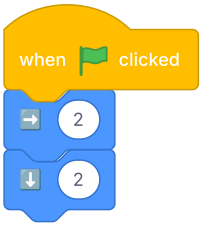
Exit ticket
What do you do after fixing a bug?

Homework
Play spot-the-difference with someone.

Answer key — Kitabash activities

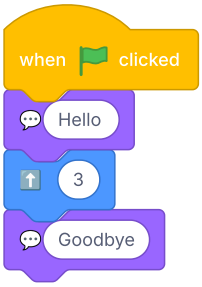
1. Lost duck (perform · block code)

One solution:



2. Grumpy goodbye (perform · block code)

One solution:

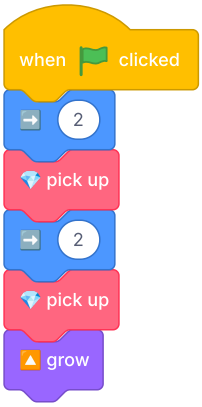


3. Keep going (perform · multiple choice)

Answer: Run it and look for the next bug

4. Busy bee (probe · block code)

One solution:



Answer key — worksheet

1. **Key words:** detective = Someone who finds clues. · test = Run it to check. · persist = Keep trying.

2. **Keep going:** A. Run it and look for the next bug — There may be more than one bug. Keep testing.
3. **Find two bugs:** ➡️ 2 should be ➡️ 3; ⬇️ should be ⬆️.

Kitabash original curriculum.